Adaptive Expert Routing for Community Load Tracking: Stability Gains and Failure Modes

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Abstract

Grouped parameter sharing makes multi-agent reinforcement learning (MARL) tractable for heterogeneous building portfolios, but permanently assigning a building to one policy actor ignores that its flexibility changes with state. We study this restriction on the 25-home IEA EBC Annex 96 Common Exercise 1 load-tracking task in CityLearn. We introduce a state-conditioned Soft Router over five complete, communication-aware MAPPO actor experts. A staged procedure first trains the router while freezing pretrained experts, then freezes the router and adapts only action heads. Across three router seeds initialized from one fixed-expert checkpoint, the resulting controller obtains $45.12 \pm 0.35\%$ CV-RMSE, $3.78 \pm 0.44\,^{\circ}$C h per building-day, and $17.30 \pm 1.33\%$ comfort exceedance. Relative to its fixed checkpoint, routing improves mean comfort severity by 16.4% while keeping CV-RMSE within 0.5 percentage points. The evaluation also exposes limits hidden by aggregate metrics: the worst-building violation duration rises from 493 to 569 ± 17 hours, joint router–expert adaptation concentrates 51.7% of traffic on one expert, and one branch becomes almost unused even under the stable schedule. These results position adaptive routing as a useful but incomplete enhancement to fixed grouping and illustrate why portfolio-level accuracy alone is insufficient evidence for grid-interactive control.

1 Introduction

Aggregators can coordinate batteries and heating, ventilation, and air conditioning (HVAC) across a community to track a portfolio-level reference load. The learning problem is intrinsically multi-agent: each home has distinct demand, thermal state, and flexibility, while success is measured at both the portfolio and building levels. CityLearn provides a reproducible environment for this setting [Vazquez-Canteli et al., 2020, Nweye et al., 2024], and recent work has shown the promise of MARL for residential flexibility [Charbonnier et al., 2025, Wilk et al., 2024]. Yet a component that improves mean load tracking can still create unacceptable local comfort violations, and a policy that is stable in one seasonal regime may fail under another. This makes evaluation design as important as aggregate reward.

Our starting point is grouped MAPPO [Yu et al., 2022]: buildings are clustered from physical and demand features, and members of a cluster share an actor. Grouping reduces the number of independent policies and encourages transfer, but creates a permanent building-to-actor assignment. Earlier experiments in this project established that communication and grouping are not neutral implementation details. Replacing the original TarMAC fusion [Das et al., 2019] with a compressed local projection and residual update reduced Vermont CV-RMSE from 49.65% to 47.01% in a matched run, while changing the clustering method or feature set materially altered both tracking and comfort. A compact feature set—battery capacity, mean heating demand, and mean non-shiftable load (3fA)—was the most stable fixed-group compromise. Our router brings the conditional-computation idea of mixture-of-experts models [Puigcerver et al., 2024, Obando Ceron et al., 2024] to this grouped control setting, where each expert is a complete closed-loop policy rather than a feed-forward sublayer.

We ask whether actor selection should remain fixed after this useful prior has been learned. Our contributions are:

- a Soft Router that selects a *complete* encoder–communication–head expert for every building and time step, avoiding incompatible latent spaces;
- a staged PPO training schedule with a fixed-group prior, entropy and load-balancing regularization, router-only learning, and low-rate head-only adaptation; and
- an evaluation that reports multi-seed mean performance, system-level effects, worst-building comfort, routing concentration, unsuccessful router variants, and a separately trained cross-climate stress test.

The main conclusion is deliberately qualified: stable routing improves the mean tracking–comfort trade-off, but does not remove dependence on strong pretrained groups and does not guarantee good tail comfort.

2 Method

Task and reward. At hour t, $N = 25$ homes jointly track reference power r_t. Let y_t be aggregate net load, $e_t = y_t - r_t$, and let $d_{i,t}$ be home i's temperature distance outside the seasonal comfort interval. Training uses the running normalized bias and tracking terms

$$B_t = \left| \frac{\sum_{\tau \le t} e_\tau}{\sum_{\tau \le t} r_\tau} \right|, \qquad E_t = \frac{\sqrt{t^{-1} \sum_{\tau \le t} e_\tau^2}}{t^{-1} \sum_{\tau \le t} r_\tau}, \tag{1}$$

and a per-building reward

$$R_{i,t} = -\left[\frac{w_B B_t + w_E E_t}{N} + w_C \left(w_I \mathbb{1}[d_{i,t} > 0] + w_d d_{i,t} \right) \right]. \tag{2}$$

The shared terms promote portfolio tracking; the local term exposes comfort costs that an aggregate objective can otherwise hide.

Fixed experts and communication. We standardize 3fA over the training homes and use Ward agglomerative clustering [Ward, 1963] to construct five actor groups. Each expert follows the centralized-training/decentralized-execution MAPPO backbone. Before actions are formed, a one-round global TarMAC block communicates across all buildings. For hidden state h_i, queries, keys, and values yield $c_i = \sum_j \mathrm{softmax}_j(q_i^\top k_j / \sqrt{d_k}) v_j$. The hybrid fusion compresses the 256-dimensional local state to 64 dimensions, combines it with the 64-dimensional message, and adds a residual update to the original state. The fixed 3fA actors are trained for 500 episodes and provide the router's expert checkpoint.

State-conditioned full-expert routing. For building i, the originally assigned encoder and global communication module produce a 256-dimensional summary h_{it}. We append seven physical quantities: normalized battery capacity and HVAC rating, state of charge (SOC), available charging space, available discharge energy, non-shiftable load, and heating demand. A shared $263 \to 64 \to 5$ network gives

$$d_{it} = \mathrm{softmax}(f_\theta([h_{it}; q_{it}])/\tau), \qquad g_{it} = (1 - \alpha) p_i + \alpha d_{it}, \tag{3}$$

where p_i is the one-hot fixed assignment. During training an expert is sampled from g_{it}; deterministic evaluation uses its largest component. Crucially, routing switches the matched encoder, communication branch, and action head together. Switching only heads can feed them representations from encoders on which they were never trained.

If expert k defines action distribution π_{ϕ_k}, the routed policy is the mixture

$$\pi_{\theta,\phi}(a_{it} \mid s_t) = \sum_{k=1}^{5} g_{itk}(\theta) \pi_{\phi_k}(a_{it} \mid s_t). \tag{4}$$

Controller	Runs	CV-RMSE (%)	|NMBE| (%)	DH_{bd}	Comfort (%)	Worst home (h)
Fixed 3fA family	5	49.86 ± 3.65	0.81 ± 0.68	4.64 ± 1.27	19.21 ± 2.85	471 ± 68
Fixed seed-42 anchor	1	44.62	0.57	4.52	20.07	493
Stable full-expert router	3	45.12 ± 0.35	1.35 ± 0.33	$\mathbf{3.78 \pm 0.44}$	$\mathbf{17.30 \pm 1.33}$	569 ± 17

Table 1: Vermont February test results. "Worst home" is the largest number of comfort-violation hours among the 25 buildings. Router runs share the same fixed seed-42 expert checkpoint.

Thus PPO [Schulman et al., 2017] can update θ even when ϕ is frozen: the likelihood of an executed action still depends on the gate. We add policy entropy and $\mathcal{L}_{\mathrm{bal}} = 5^{-1} \sum_k (\bar{g}_k - \bar{p}_k)^2$, where $\bar{p}$ is the original group proportion.

Stable schedule. The router uses $\tau = 0.5$, $\alpha : 0.5 \to 1$ over 150 episodes, learning rate 10^{-4}, entropy scale 0.05, and balance coefficient 0.01. For episodes 1–500 all expert parameters are frozen. For episodes 501–1000 the router, encoders, and communication module are frozen and only action heads adapt at 10^{-5}. This removes the moving target created when both router semantics and expert behavior change simultaneously.

3 Evaluation

Protocol and metrics. Vermont uses January for training and February for testing, at one-hour resolution; the comfort interval is 20–24°C. We report CV-RMSE, absolute NMBE, comfort-exceedance rate, and comfort severity $\mathrm{DH}_{bd} = \sum_{i,t} d_{i,t} \Delta t / (N D_{\mathrm{test}})$. All are lower-is-better. Five independent seeds summarize the fixed 3fA family. The final router uses three router-training seeds (0, 1, 42), but all start from the same seed-42 fixed checkpoint. Table 1 therefore includes both the multi-seed family and the exact anchor; the routed and five-seed rows should not be interpreted as fully independent end-to-end seed comparisons.

Mean and system-level effects. Against its exact fixed anchor, the stable router reduces degree-hours by 16.4% and comfort-exceedance rate by 13.8%, while CV-RMSE increases only 0.50 percentage points. Across the broader fixed family, routing also has a lower mean CV-RMSE, but this comparison benefits from starting all routers from the strong seed-42 checkpoint. Absolute bias worsens from 0.57% at the anchor to $1.35 \pm 0.33\%$, exposing a real tracking–comfort trade-off. Downstream grid metrics are mixed: the router's mean monthly peak falls from 281.9 to 276.2 kW and ramping from 487.2 to 476.5 kW, while site energy rises from 70.76 to 72.36 MWh. Load-share fairness is essentially unchanged (Gini 0.1438 versus 0.1435 ± 0.0126).

Ablations. The progression in Figure 1 shows that a good router cannot be obtained merely by adding a gate. End-to-end training has acceptable CV-RMSE (47.60%) but poor comfort severity (10.02 °C h/building-day). A shared-encoder three-stage model is worse still (52.44% CV-RMSE and 31.35% comfort exceedance). Preserving complete expert branches improves CV-RMSE to 45.45%; freezing the learned router and restricting subsequent adaptation to the heads gives the final stable result. Full numerical ablations are in Appendix A.

4 Failure modes, regime shift, and limitations

Routing collapse. After 500 router-only episodes, the largest expert share is 36.4%. Continuing joint full-expert adaptation reduces gate entropy from 0.275 to 0.171 and sends 51.7% of decisions to one expert. Freezing the router during head adaptation preserves entropy at 0.276 and limits the largest share to 37.0%. Nevertheless, one expert receives only 0.015–0.036% of final traffic across the three stable runs. Entropy and balance penalties prevent complete one-expert collapse, not expert under-use.

Aggregate gains hide local risk. The stable router improves average comfort but raises the worst-building violation duration from 493 hours at its anchor to 569 ± 17 hours. The five-seed fixed family averages 471 ± 68 hours. This reversal is invisible in CV-RMSE and portfolio mean comfort

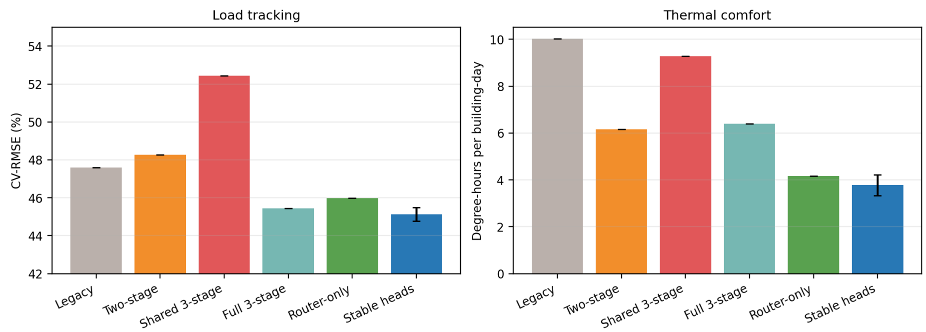

Figure 1: Soft-router development stages. Lower is better on every panel; the final bar reports the three-seed stable-head mean.

and is the strongest reason not to interpret the method as deployment-ready. Explicit per-building constraints or a tail-aware objective are needed before narrow autonomy.

Climate stress test. We separately retrain the 3fA design in Texas, replacing mean heating with mean cooling demand and using August/September train/test months. It reduces CV-RMSE to $78.83 \pm 4.90\%$, versus 163.74% for no control and 207.63% for the rule-based controller, but retains a large $45.57 \pm 1.98\%$ absolute bias. This tests transfer of the *feature design*, not zero-shot transfer of a Vermont policy; it therefore does not establish out-of-distribution policy robustness.

Scope. All final router seeds reuse one unusually strong expert checkpoint, the study contains only 25-building Vermont and Texas simulations, and CityLearn enforces asset dynamics but contains no distribution-network or AC power-flow model. Consequently we make no claim about voltage, line-flow, or contingency feasibility. Future work should train complete expert–router systems from independent initializations, evaluate unseen building portfolios and target profiles, optimize tail comfort, and couple control to AC network constraints.

5 Conclusion

State-conditioned routing can relax a permanent building-to-actor assignment without discarding useful grouped policies. The effective recipe is conservative: route complete experts, learn selection against frozen reference policies, and adapt only small output heads after the gate stabilizes. Its benefits are meaningful at the portfolio mean, but the worst-building result and residual expert under-use show why component accuracy is insufficient. For grid-interactive MARL, router concentration and per-asset tails should be reported alongside aggregate tracking metrics.

Generative AI disclosure

A generative-AI system was used to assist with manuscript organization and language editing. The authors verified the technical descriptions, numerical results, and references against the implementation, recorded experiment artifacts, and cited sources. Generative AI was not part of the research method and did not generate experimental data or figures.

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A Router ablations and additional protocol

Variant	Main change	CV-RMSE (%)	|NMBE| (%)	DH_{bd} (°C h)	Comfort (%)
Legacy end-to-end	random policies, joint training	47.60	4.31	10.02	24.64
Two-stage freeze200	pretrained experts	48.26	1.35	6.15	21.76
Two-stage prior0.7	softer prior	46.54	1.02	6.60	22.05
Two-stage no capacity	remove seven router inputs	46.94	0.57	5.53	19.54
Three-stage shared	shared encoder	52.44	7.48	9.27	31.35
Three-stage full expert	preserve complete branches	45.45	2.28	6.39	20.42
Stable router-only	experts always frozen	45.99	1.22	4.16	19.17
Stable heads (3 seeds)	router freeze + head adaptation	45.12 ± 0.35	1.35 ± 0.33	3.78 ± 0.44	17.30 ± 1.33

Table 2: Development ablation. Single-run rows use seed 42; only the final row reports multiple router-training seeds.

The MAPPO configuration uses actor and critic learning rates 3×10^{-4}, $\gamma = 0.99$, GAE $\lambda = 0.95$, PPO clip 0.2, 10 PPO epochs, four mini-batches, and 256-dimensional two-layer actors. The communication block uses 32-dimensional queries/keys, 64-dimensional values, one global round, and a residual update. The fixed actors and final router schedules each see complete one-month training episodes. Evaluation is deterministic over the held-out following month. The Vermont comfort interval is 20–24°C; Texas uses 22–26°C.

B Cross-climate feature-design stress test

Climate/controller	CV-RMSE (%)	$\lvert$NMBE$\rvert$ (%)	DH_{bd} ($^{\circ}$C h)	Comfort (%)
Vermont 3fA	49.64 ± 1.59	3.04 ± 1.49	5.10 ± 1.95	18.74 ± 1.90
Texas 3fA	78.83 ± 4.90	45.57 ± 1.98	1.59 ± 0.31	11.22 ± 1.66
Texas no control	163.74	79.12	–	–
Texas rule based	207.63	89.57	–	–

Table 3: Three-seed 3fA role-based comparison. Vermont uses capacity–heating mean–non-shiftable-load mean; Texas uses capacity–cooling mean–non-shiftable-load mean and is retrained from scratch. Climate scales differ, so Vermont and Texas rows should not be directly ranked.

AI4PowerGrids domain checklist

1. **Physics feasibility.** *Not applicable to AC power-flow feasibility.* CityLearn enforces the simulated battery, HVAC, and building thermal dynamics and bounds the control actions, while thermal violations are reported explicitly in Sections 3–4. The benchmark does not include a distribution-network model; consequently, voltage, line-flow, and full AC power-flow feasibility are not evaluated and no such claim is made.
2. **Out-of-distribution evaluation.** Section 4 and Appendix B evaluate the same feature-design principle in two seasonal/climate regimes (Vermont heating and Texas cooling), each with a chronological held-out month. The Texas controller is retrained, so this is not zero-shot evaluation of one policy on an unseen climate or topology. This missing test is stated as a limitation.
3. **Failure modes.** Section 4 reports routing concentration, an almost unused expert, failed end-to-end and shared-encoder variants, large Texas bias, and the reversal between average and worst-building comfort. Appendix A gives the numerical ablation results.
4. **System-level effect.** Section 3 reports portfolio CV-RMSE, NMBE, peak demand, total energy, ramping, and load-share fairness, together with building-level comfort. This evaluates the routed policy inside the complete closed-loop CityLearn portfolio rather than as a standalone classifier.
5. **Data and code.** The experiments use the open CityLearn implementation and IEA EBC Annex 96 Common Exercise 1 benchmark data. We plan to provide the training and evaluation code, environment specifications, configuration scripts, processed metric tables, and plotting code as anonymized supplementary material during review, followed by a de-anonymized public repository upon acceptance.